The Functional Equation

The Symmetry the BN Flow Inherits

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Abstract

The functional equation $\zeta(s) = \chi(s)\zeta(1-s)$ is a symmetry of ζ under $s \mapsto 1-s$, with $\chi(s) = 2^s\pi^{s-1}\sin(\pi s/2)\Gamma(1-s)$. On the critical line $\mathrm{Re}(s) = 1/2$: $|\chi(1/2+it)| = 1$, so $\zeta(1/2+it) = \chi(1/2+it)\overline{\zeta(1/2+it)}$ — ζ is its own reflection. The BN flow inherits this symmetry: the optimal polynomial $P_N^*(s)$ satisfies $P_N^*(1-s) \approx P_N^*(s)/\chi(s)$ in the large-N limit. RH says the zeros are exactly on the axis of this symmetry; the functional equation is the constraint that forces them there or allows them to escape in conjugate pairs.

1 The Functional Equation

Definition 1 (Completed zeta function)**.** The completed zeta function is:

$$\xi(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

It satisfies the exact symmetry $\xi(s) = \xi(1-s)$ and is entire with simple zeros exactly at the nontrivial zeros of ζ. The functional equation $\zeta(s) = \chi(s)\zeta(1-s)$ is equivalent, with $\chi(s) = \pi^{s-1/2}\Gamma((1-s)/2)/\Gamma(s/2)$.

Proposition 2 (Zeros come in symmetric pairs)**.** *If $\rho = \sigma + i\gamma$ is a nontrivial zero of ζ, so are $\bar{\rho} = \sigma - i\gamma$ (reality of coefficients) and $1-\rho = (1-\sigma)+i\gamma$ (functional equation). Under RH ($\sigma = 1/2$): all four coincide pairwise — $\bar{\rho} = 1-\rho$ and the orbit has size 2: $\{1/2+i\gamma, 1/2-i\gamma\}$. Under ¬RH ($\sigma \neq 1/2$): the orbit has size 4: $\{\sigma \pm i\gamma, (1-\sigma)\pm i\gamma\}$. Each off-critical zero ρ_0 comes with a partner $1-\rho_0$ on the opposite side of the critical line.*

Remark 3 (The pairing and the residual). From [1]: the permanent residual is $d_\infty^2 = \sum_\rho c_\rho/|\rho|^2|\rho-1|^2$. The pairing $\rho \leftrightarrow 1-\rho$ gives: $c_\rho/|\rho|^2|\rho-1|^2 = c_{1-\rho}/|1-\rho|^2|\rho|^2$ (the two terms are equal if $c_\rho = c_{1-\rho}$). Each off-critical pair $\{\rho_0, 1-\rho_0\}$ contributes $2c/|\rho_0|^2|\rho_0-1|^2 > 0$ to d_∞^2. Under RH, there are no pairs; $d_\infty^2 = 0$.

2 The Symmetry of the BN Flow

Definition 4 (BN reflection operator)**.** Define the reflection R on $L^2(0,1)$ by $(Rf)(x) = f(x)/x$ (the Mellin reflection $s \mapsto 1-s$ in L^2 coordinates). Then $R\rho_{1/k}(x) = \rho_{1/k}(x)/x$ has

Mellin transform $\widehat{R\rho_{1/k}}(s) = \widehat{\rho}_{1/k}(1-s) = (k^{s-1} - k^{-1})\zeta(1-s)/(1-s)$. Under the functional equation: $(k^{s-1} - k^{-1})\zeta(1-s)/(1-s) = \chi(1-s)(k^{s-1} - k^{-1})\zeta(s)/(1-s)$.

Proposition 5 (Symmetry of the Gram matrix)**.** *The BN Gram matrix satisfies $G_{kl} = G_{lk}$ (symmetric), but it does not satisfy $G_{kl} = G_{\sigma(k),\sigma(l)}$ for any non-trivial permutation σ of the integers. The functional equation does not induce a symmetry of the Gram matrix directly — but it does induce a symmetry of the optimal polynomial P_N^* in the limit.*

Theorem 6 (Functional equation for P_N^*)**.** *Let $P_N^*(s) = \sum_{k=2}^{N+1} a_k^*(k^{-s} - k^{-1})$ be the BN optimal polynomial. Then in the L^2 sense on $\mathrm{Re}(s) = 1/2$:*

$$\lim_{N\to\infty} \left\| P_N^*(s)\zeta(s) - 1 \right\|_{L^2(1/2)} = d_\infty,$$

and under RH ($d_\infty = 0$): the functional equation implies that $P_\infty^*(s)\zeta(s) = 1$ satisfies

$$P_\infty^*(1-s) = P_\infty^*(s)/\chi(s),$$

i.e., the limiting polynomial transforms under $s \mapsto 1-s$ by the factor $\chi(s)^{-1} = \zeta(1-s)/\zeta(s)$.

Proof sketch. If $P_\infty^*(s)\zeta(s) = 1$ (achieved under RH at $d_\infty^2 = 0$), then $P_\infty^*(1-s)\zeta(1-s) = 1$. By the functional equation $\zeta(1-s) = \zeta(s)/\chi(s)$: $P_\infty^*(1-s) \cdot \zeta(s)/\chi(s) = 1$, so $P_\infty^*(1-s) = \chi(s)/\zeta(s) \cdot \chi(s) = \chi(s)P_\infty^*(s)$. Wait: $P_\infty^*(s)\zeta(s) = 1 \Rightarrow P_\infty^*(s) = 1/\zeta(s)$. And $P_\infty^*(1-s) = 1/\zeta(1-s) = \chi(s)/\zeta(s) = \chi(s)P_\infty^*(s)$. □

Remark 7 (The reflection law). The limiting BN polynomial $P_\infty^*(s) = 1/\zeta(s)$ satisfies $P_\infty^*(1-s) = \chi(s)P_\infty^*(s)$. This is the reflection law of $1/\zeta$: the reciprocal inherits the functional equation. At $s = 1/2 + it$ on the critical line: $P_\infty^*(1/2 - it) = \chi(1/2 + it) \cdot P_\infty^*(1/2 + it)$. Since $|\chi(1/2 + it)| = 1$, the reflection is a pure phase rotation: $|P_\infty^*(1/2 - it)| = |P_\infty^*(1/2 + it)|$. The modulus of $1/\zeta$ is symmetric about the real axis on the critical line.

3 The Functional Equation and the Zeros

Theorem 8 (Pairing forces the wall to be symmetric)**.** *Under $\neg$RH, each off-critical zero $\rho_0 = \sigma_0 + i\gamma_0$ contributes to d_∞^2 via both ρ_0 and its partner $1 - \rho_0$. The pair contributes:*

$$d_\infty^2 \geq \frac{c}{|\rho_0|^2|\rho_0 - 1|^2} + \frac{c}{|1-\rho_0|^2|\rho_0|^2} = \frac{2c}{|\rho_0|^2|\rho_0 - 1|^2}.$$

The wall is doubled by the functional equation: every off-critical zero brings its partner.

Theorem 9 (The functional equation pins zeros to the line)**.** *Consider the family of distributions of zeros consistent with the functional equation: each off-critical zero ρ_0 must have a partner $1 - \rho_0$. As $\sigma_0 \to 1/2^+$ (zero approaching the critical line from the right), both ρ_0 and $1 - \rho_0 = (1 - \sigma_0) + i\gamma_0$ approach $1/2 + i\gamma_0$. In the limit $\sigma_0 = 1/2$: the pair merges to a single point on the line.*

The functional equation does not force $\sigma_0 = 1/2$; it only requires ρ_0 and $1 - \rho_0$ to exist together. But it constrains the permanent residual: $d_\infty^2(\sigma_0) = 2c/|\rho_0|^2|\rho_0 - 1|^2$, and $\lim_{\sigma_0 \to 1/2} d_\infty^2(\sigma_0) = ?$.

At $\sigma_0 = 1/2$: the zero is on the line; $d_\infty^2 = 0$. For any $\sigma_0 > 1/2$: $d_\infty^2 \geq 2c/\gamma_0^2 > 0$. The transition is discontinuous: d_∞^2 jumps from 0 to > 0 at $\sigma_0 = 1/2$. This jump IS the wall.

Remark 10 (Why the functional equation is not enough). Theorem 9 shows the wall is a discontinuity in d^2_∞ at $\sigma_0 = 1/2$. The functional equation requires that zeros pair as $\{\rho_0, 1-\rho_0\}$, which pushes toward the critical line but does not reach it: ρ_0 and $1-\rho_0$ are symmetric about 1/2, and their midpoint is 1/2, but the zeros themselves are at $\sigma_0 \neq 1/2$.

To force $\sigma_0 = 1/2$, one needs an additional input. In the BN language: $d^2_\infty = 0$, which is the content of RH. The functional equation says: zeros come in pairs centered at 1/2. RH says: the pairs are degenerate (zero-width). The functional equation is necessary but not sufficient for RH.

4 The Symmetry of the BN Distance

Proposition 11 (Symmetry of the spectral density). *The spectral density* $\sigma(t) = |\zeta(1/2+it)|^2/(1/4+t^2)$ *satisfies:* $\sigma(t) = \sigma(-t)$ *(even, from* $\overline{\zeta(1/2+it)} = \zeta(1/2-it)$*). The functional equation adds:* $|\zeta(1/2+it)|^2 = |\chi(1/2+it)|^2|\zeta(1/2-it)|^2 = |\zeta(1/2-it)|^2$*. So* σ *is even and self-consistent under the functional equation.*

Theorem 12 (BN distance inherits the symmetry). *The BN optimal polynomial* $P^*_N(s)$*, when evaluated at conjugate points, satisfies* $P^*_N(1/2-it) = \overline{P^*_N(1/2+it)}$ *(by reality of the coefficients* a^*_k*). Combined with Theorem 6 in the limit* $N \to \infty$ *under RH: the BN flow converges to the unique solution* $P^*_\infty = 1/\zeta$ *that satisfies both the reflection law and the functional equation. The symmetry* $\sigma(t) = \sigma(-t)$ *means the BN Gram matrix* G_N *is centrosymmetric ([3]):* $G_N = JG_NJ$ *where* J *is the reversal permutation matrix. This centrosymmetry is an unconditional consequence of the functional equation.*

References

[1] Rincón, D. (2026). The Density. Preprint.

[2] Rincón, D. (2026). The Explicit Formula. Preprint.

[3] Rincón, D. (2026). The Gram Matrix. Preprint.

[4] Rincón, D. (2026). Widom. Preprint.

[5] Rincón, D. (2026). Wisdom. Preprint.

[6] Rincón, D. (2026). Möbius. Preprint.